Kuratowski Calculus Engine

Unified Code Manual

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Part of Modulo Synthesis / Fractal Entropic Geometrodynamics

"The universe is strictly computable, finite, and structurally homeostatic. The rest is just counting."

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CHAPTER 1

Theory

Chapter Overview

- *The two axioms of Fractal Entropic Geometrodynamics.*
- *The Sole Principle of Interaction (microscopic law).*
- *Kuratowski's Theorem and the origin of matter, forces, and generations.*
- *The Three Geometries: Order, Matter, and Logic.*
- *How each physics concept maps to a code module and data structure.*

1.1 Axiom 1 — Causal Sets

Spacetime is not a continuous manifold but a **discrete partial order** — a directed acyclic graph (DAG) where each node is a spacetime event and each edge encodes causal influence: $A \to B$ means "A can causally influence B." The DAG is generated by Poisson sprinkling N points into a 4D causal diamond ($|t| + r \leq T/2$) with volume $V = \pi T^4/24$ and $T = (24N/\pi)^{1/4}$ giving fundamental density $\rho \approx 1$.

Axiom 1: Causal Sets (from Vol I)

Spacetime is a locally finite partially ordered set. The order relation encodes causal structure; the counting measure (number of elements in an Alexandrov set) encodes spacetime volume. All geometry is emergent from order + number.

1.2 Axiom 2 — Quantum Dynamics

Axiom 2: Quantum Dynamics: The Non-Commutative Postulate (from Vol II)

The geometry and quantum phases of the causal set are evaluated over a **non-commutative von Neumann algebra**. Macroscopic geometry is not fundamental; it is an emergent phenomenon dictated by the quantum Fisher Information Metric (Petz metric) defined over the density states of causal histories. Causal histories do not commute: if events A and B are spacelike separated, the order in which they are evaluated affects the quantum phase.

Together, Chentsov's Theorem (classical, macroscopic) and Petz's Theorem (quantum, microscopic) establish a single information-geometric foundation for all of physics. Gravity is the coarse-grained Fisher geometry at large N; quantum mechanics is the fine-grained Petz geometry at small N.

1.3 The Sole Principle of Interaction

Axiom 3: The Sole Principle of Interaction (Microscopic Law)

The entirety of microscopic network dynamics is governed by a single mandate: **the preservation of unitarity** (probability conservation) across discrete, non-commutative causal links:

$$\sum_{\text{paths}} P(i \to f) = 1$$

All emergent forces (gauge fields) and geometric structures (gravity) are **homeostatic mechanisms** enacted by the causal network to satisfy this unitary principle under the topological constraints of Kuratowski's Theorem.

1.4 Hasse Diagram

The **transitive reduction** of the causal DAG. If $A \to B \to C$, the direct edge $A \to C$ is redundant and is removed. The Hasse diagram retains only nearest-neighbour causal links — the skeleton of spacetime.

1.5 Causal Prism — The Fundamental Particle

CRITICAL: The transitive reduction in Phase 1 produces a *triangle-free* DAG. K_4 complete subgraphs (4 mutually connected events) **cannot exist** in such a graph — transitive reduction eliminates all triangles by definition. The correct fundamental particle is the **Causal Prism** ($K_{2,N}$ bipartite, $N \geq 3$).

A Causal Prism consists of:

- Two **poles**: past pole u (origin) and future pole v (destination)
- $N \geq 3$ **belly nodes** (intermediates w_i)
- Directed edges: $u \to w_i$ and $w_i \to v$ for all $i \in \{1 \dots N\}$
- The belly = children$(u) \cap$ parents(v)

These topological defects:

- Anchor probability flow and define particle-like spectral signatures.
- Cannot be "untied" without breaking the graph's connectivity — they are topologically stable.
- Are the *maximal* bipartite structures permitted by the triangle-free constraint.

1.6 Threat Contraction — Colour Confinement

Theorem 4: Kuratowski's Theorem (The Topological UV Limit)

A finite graph G is planar if and only if it does not contain a topological minor of K_5 (the complete graph on 5 vertices) or $K_{3,3}$ (the complete bipartite graph on $3+3$ vertices):

$$K_5 \not\preceq_T G_{\text{UV}} \quad \text{and} \quad K_{3,3} \not\preceq_T G_{\text{UV}}$$

This is the absolute UV boundary of the theory.

When an external node is connected to **both poles** of a Causal Prism AND ≥ 2 intermediates, it threatens the topological stability and is **absorbed** (vertex contraction) into the higher-degree pole — a discrete analogue of colour confinement. The topology enforces that quarks cannot exist in isolation.

Key Result

The Strong Force is not a fundamental gauge field. It is a strict topological obstruction — the causal network's response to the Kuratowski UV limit — preventing forbidden topological minors in the ultraviolet.

1.7 Spectral Dimension — Gravity

The spectral dimension measures how quickly a random walker returns to its starting point after t steps:

$$d_S(t) = -2\,\frac{d(\ln P)}{d(\ln t)}$$

On a smooth 4D manifold, $d_S \to 4$. On the Hasse graph:
- **Vacuum:** d_S diverges (boundary saturation of the unstructured bulk).
- **With Causal Prism matter:** d_S stabilises at a finite value — the topological knot traps probability flow.

1.8 Causal Flux — Electromagnetism

Directed walkers follow the arrow of time (cause → effect only). The transmission probability between particle generations measures electromagnetic-like interactions:
- **Attraction:** Gen1 → AntiGen1 (opposite charges).
- **Repulsion:** Gen1 → Gen1 (same charges).

1.9 Three Generations — The Kuratowski Limit

The Standard Model has three copies of matter (electron/muon/tau, up/charm/top). In FEG the answer is a strict topological obstruction:
1. UV gravity compresses $d_S \to 2$ (effectively 2D topology).
2. Kuratowski's Theorem forbids K_5 in planar graphs.
3. The transitive reduction produces a triangle-free DAG.
4. Causal Prisms ($K_{2,N}$ bipartite) are the fundamental particles.

Causal Prisms are classified by the **Vol I phase function** $\varphi(w) = \text{sign}(\text{bulk\_momentum}[w])$. For each prism P with intermediates $\{w_1, \ldots, w_N\}$:
- $g(P) = |\{\varphi(w_i)\}|$ counts distinct phase values — the **generation number** (1, 2, or 3).
- $\Phi(P) = \sum_i \varphi(w_i)$ gives the **net phase** — matter if $\Phi \geq 0$, antimatter if $\Phi < 0$.

Gen1 = g=1 ∧ Φ≥0, Gen2 = g=2, Gen3 = g=3, Anti1 = g=1 ∧ Φ<0. The constraint $g \leq 3$ (at most 3 distinct sign values: −1, 0, +1) enforces exactly 3 stable generations.

1.10 The Three Geometries

The theory rests on three geometric pillars, each discovered through the simulation engine:

Key Result

The Three Geometries of Fractal Entropic Geometrodynamics:
1. ***Geometry of Order*** *(Vol I): Transitive Reduction → Lorentz invariance. The Hasse diagram encodes the causal structure of spacetime; counting elements in an Alexandrov set reproduces the volume element.* Code: `diamond.rs` *builds the Hasse DAG.*
2. ***Geometry of Matter*** *(Vol II): Causal Prism* $K_{2,N}$ *→ mass hierarchy. Topological defects in the Hasse graph carry quantized topological mass* N *and are classified into exactly three generations by the Kuratowski limit.* Code: `skyrmion.rs` *detects prisms and classifies generations.*
3. ***Geometry of Logic****:* $O(N)$ *scaling ⇔ physical locality. Bounded Hasse degree* $(D \leq 15)$ *implies that every topological operation—prism detection, threat contraction, spectral walk—is local, yielding* ***provably linear*** $O(N)$ *computational complexity. This is not an engineering optimisation but a* computational proof *that the theory encodes physical locality as an information principle.* Code: *2-hop*

candidate search in `skyrmion.rs`; single-pass streaming in `diamond.rs`.

Theorem 5: $O(N)$ Locality Theorem

Let $\mathcal{H}$ be the Hasse diagram of N events sprinkled into a 4D causal diamond at density $\rho \approx 1$. Then:

1. Every Hasse link has bounded proper time: $\tau_{\text{link}} \leq 8$.
2. Hence every node has bounded Hasse degree: $D \leq D_{\max} \approx 15$.
3. Causal Prism detection via 2-hop traversal is $O(N \cdot D_{\max}^3) = O(N)$.
4. All other phases (edge generation, spectral walkers, output) are already $O(N)$ or $O(W \cdot t)$.

The entire simulation scales linearly with the number of spacetime events.

1.11 Physics → Code Dictionary

Physics Concept	Code Module	Data Structure
Spacetime	`diamond.rs`	DAG of `[f64; 4]` points
Causal link	`diamond.rs`	CSR edge (`adj_head, adj_data`)
Particle (matter)	`skyrmion.rs`	Causal Prism ($K_{2,N}$ bipartite)
Strong force	`skyrmion.rs`	Threat contraction `merge_into[i]`
Charge (bulk momentum)	`diamond.rs`	`bulk_momentum: Vec<i32>`
Particle generation	`skyrmion.rs`	Vol I phase: $g(P) = \lvert\{\varphi(w_i)\}\rvert$
Antimatter	`skyrmion.rs`	CPT conjugate: $g{=}1 \wedge \Phi(P){<}0$
Gravity (geometry)	`spectral.rs`	Undirected walkers → $P(t) \to d_S(t)$
Electromagnetism	`spectral.rs`	Directed walkers → transmission prob.
Locality ($O(N)$)	`skyrmion.rs`	2-hop traversal, bounded degree $D \leq 15$
Topology export	`anim_export.rs`	`TopologySlice` → bincode `.slice` file

Table 1.1: Mapping between physics concepts and code implementations.

CHAPTER 2

Architecture

Chapter Overview

- *The four-phase pipeline: sprinkle → hasse → defect → walkers → csv.*
- *Two execution paths: in-memory vs streaming (single-pass $O(N)$).*
- *Tier system: eigendecomposition vs Monte Carlo.*

2.1 Data Flow

```
sprinkle(N) -> [f64;4] points
     |
build_hasse() -> CSR adjacency (adj_head, adj_data) + bulk_momentum
     |
apply_defect() -> Causal Prisms, K5 threat absorption, generation classification
     |
    +--[--export-slice]--> write_slice() -> .slice file (bincode, early exit)
     |
run_walkers() -> P(t), dS(t) for vacuum/defect/generations/flux
     |
write_csv() -> results.csv
```

2.2 Phase Diagram

Phase	Module	Physics
1	`diamond.rs`	Vacuum generation: Poisson sprinkling + Hasse diagram
2	`skyrmion.rs`	Kuratowski contraction: Causal Prism detection + K_5 absorption
3	`spectral.rs`	Spectral dimension: random walk return probability
4	`output.rs`	CSV serialisation of ensemble-averaged observables

Table 2.1: The four simulation phases.

2.3 Execution Paths

2.3.1 In-Memory

Selected by `--inmemory` flag or auto-selected when RAM is sufficient. All data structures live in RAM. Realisations run in parallel via Rayon with an auto-selected concurrency limit (4 threads, or 2 for $N > 10$M; overridable via `--threads`). Causal Prism detection uses 2-hop local candidate search: $O(N \cdot D^3)$ with $D \leq 15$.

2.3.2 Streaming

Selected by `--stream` flag or auto-selected when RAM is insufficient. Phase 1 uses `scan_edges_with_analysis()` to perform a **single-pass** scan: degree counting, core classification, and core-edge collection are merged into one streaming pass over the edge generator. Causal Prism detection then operates on the core-induced CSR subgraph using the same 2-hop local search as the in-memory path. RAM usage stays under ∼1.3 GB per realisation even at $N = 100$M. Concurrency is auto-selected by CPU count (capped at 8; overridable via `--threads`).

Mode selection is automatic based on available system RAM (via `sysinfo`), configurable via `causal_set.toml`, or overridable with CLI flags.

2.4 Tier System

Tier	Condition	Algorithm	Complex
Eigendecomp	$N \leq 3{,}000$ (default `--eigen-cutoff`)	Dense $S = D^{-1/2}AD^{-1/2}$ eigendecomp	$O(N^3)$
Monte Carlo	$N > 3{,}000$ (default `--eigen-cutoff`)	Lazy random walkers on CSR	$O(W \cdot t)$

Table 2.2: Spectral dimension computation tiers.

CHAPTER 3

Module Reference: `diamond` — Phase 1

Vacuum Generation

- *Poisson sprinkling in a 4D causal diamond.*
- *Hasse diagram construction (transitive reduction).*
- *Two tiers: sparse A^2 reduction ($N \leq 15k$) and OCI geometric construction ($N > 15k$).*

3.1 Functions

3.1.1 `sprinkle(n, rng)` → `(Vec<[f64;4]>, f64)`

Rejection-samples n points inside $|t| + r \leq T/2$ with $T = (24N/\pi)^{1/4}$ giving density $\rho \approx 1$. The volume of the 4D causal diamond is $V = \pi T^4/24$. The diamond is the intersection of the future light cone of $(t = -T/2)$ and the past light cone of $(t = +T/2)$.

3.1.2 `build_hasse_sparse(pts)` → `(coords, adj_head, adj_data, momentum)`

Full causal closure followed by sparse A^2 reduction. Edge (i, j) is a Hasse link iff $A^2[i, j] = 0$ (no 2-hop path exists). Used for $N \leq 15{,}000$. Parallel over source nodes via Rayon.

3.1.3 `build_hasse_direct(pts)` → `(coords, adj_head, adj_data, momentum)`

Direct geometric construction using the **Occupied Cell Index (OCI)** — a spatial indexing paradigm that replaces naive geometric shell enumeration (96% empty cells) with per-time-layer lists of occupied cells, queried via binary search. Features:

- Points are quantised onto an integer lattice and sorted by cell index for cache-coherent traversal (`GridPoint` struct).
- Per-time-layer lists of occupied cells, sorted by X coordinate.
- Binary search for the active X-band within the light cone.
- **Topological cutoffs** (Phase 1 optimisations):
  - `MAX_CAUSAL_DEPTH = 4`: proper-time cutoff. Hasse links beyond $\Delta t > 4$ are exponentially suppressed at $\rho \approx 1$ (was 10).
  - `MAX_HASSE_DEGREE = 15`: degree cap (Valve 1) — a node with 15+ descendants has fully populated its causal shadow (was unbounded).
- **Alexandrov volume short-circuit (Valve 2):** If every causal candidate in a Δt-shell is transitively blocked (non-empty Alexandrov set between u and v), further shells have strictly larger Alexandrov volume and are guaranteed blocked. Breaks immediately when degree ≥ 1.
- **Adaptive dry-shell termination (Valve 3):** 2 consecutive dry Δt-shells (no new Hasse links) + ≥ 2 existing descendants triggers cutoff.
- Hollow light-cone shell optimisation: skip deep interior points where proper time > `MAX_CAUSAL_DEPTH` units.

3.1.4 `stream_edges_to_file(n_total, chunk_size, path, seed)`

Streaming variant that writes Hasse edges to disk using a sliding window, for N too large to fit in RAM. Includes the same three termination valves (degree cap, Alexandrov short-circuit, adaptive dry-shell) as the in-memory path. Output format: continuous stream of (u32, u32) in little-endian. Both physics invariants are preserved: full light cone search and transitive reduction via locally cached children coordinates.

3.1.5 sprinkle_chunk(n_expected, t_min, t_max, half_t, rng, start_idx)

Poisson-sprinkle points into a time-slab $[t_{\min}, t_{\max}]$. Used internally by the streaming path to fill the sliding window buffer.

3.1.6 scan_edges_sparse_ext(n_total, chunk_size, seed, on_batch)

Extended streaming edge scanner. Identical to `scan_edges_sparse` except the callback receives the current `safe_threshold` as a second argument: `FnMut(&[(u32, u32)], f64)`. This enables callers to track node finalization (a node's total degree is known once $\text{safe\_threshold} > t_{\text{node}} + \text{safety\_margin}$).

3.1.7 scan_edges_with_analysis(n_total, chunk_size, seed, core_num, core_den)

Single-pass streaming analysis that merges degree counting, core classification, and core-edge collection into one scan over the edge generator. Returns `(in_deg, out_deg, is_core, directed_core_edges)`.

Pipeline inside the callback:

1. Update `in_deg[v]` and `out_deg[u]` for each edge.
2. Track finalized nodes via safe-threshold delay $(\text{safe\_threshold} - 2 \cdot \text{max\_dt} - 4.0)$.
3. Running degree histogram (10,000 bins) estimates the core threshold after 5% warmup.
4. Classify finalized nodes as core or non-core.
5. Collect edges where both endpoints are classified core; buffer edges where one endpoint is pending.
6. Final exact $O(N)$ histogram pass corrects any drift ($\pm 0.5\%$) and re-checks all collected core edges.

Key Result

The single-pass architecture eliminates the two-pass bottleneck (degree counting + core-edge collection), halving wall-clock time for $N > 1M$ streaming runs. At $N = 100M$ each realisation uses $\sim$1.3 GB RAM ($13 \cdot N$ bytes).

3.2 Key Concepts

Definition 1: GridPoint

A spacetime event quantised onto an integer lattice for cache-coherent traversal. Fields: `p` (coordinates), `orig_idx`, `cell` (linearised 4D grid coordinate), `qt/qx/qy/qz` (quantised coordinates). Points are sorted by cell index so that spatially nearby events are contiguous in memory.

Definition 2: Occupied Cell Index (OCI)

Paradigm shift from geometric to spatial indexing. The naive approach iterates all cells in the light-cone shell for each source node. At large N the grid is sparse ($> 96\%$ cells empty), making geometric enumeration wasteful. OCI inverts the lookup:

1. Build per-time-layer lists of *occupied* cells only.
2. Sort each layer by X coordinate.
3. For each source, binary-search the X-band of the light cone within the occupied list, skipping all empty space.

This reduces the inner-loop cost from $O(\text{shell\_volume})$ to $O(\text{occupied\_cells})$, a $\sim 20\times$

speedup at $N > 100\text{k}$.

CHAPTER 4

Module Reference: `skyrmion` — Phase 2

Kuratowski Contraction

- *Pure integer topology — zero floating-point.*
- *Core identification → Causal Prism ($K_{2,N}$ bipartite) detection → threat contraction → generation classification.*
- *Vol I phase classification: $g(P) = |\{\varphi(w_i)\}|$ (generation), $\Phi(P) = \sum \varphi(w_i)$ (matter/antimatter).*

4.1 Algorithm

CRITICAL: The transitive reduction produces a triangle-free DAG. K_4 complete subgraphs **cannot exist**. The fundamental particle is the **Causal Prism** ($K_{2,N}$ bipartite, $N \geq 3$): two poles with $N \geq 3$ belly nodes (intermediates satisfying $u \to w_i \to v$).

1. **Core selection:** Top 10% of nodes by undirected Hasse degree ($\approx$ the diamond's combinatorial centre).
2. **Causal Prism formation ($O(N)$ forward-forward 2-hop search): Minimum valence pruning:** nodes with out-degree < `MIN_PRISM_SHARED` (3) are skipped before the 2-hop traversal, and candidate poles v with in-degree < 3 are skipped before the belly intersection. For each remaining core node u, collect 2-hop candidates by traversing $u \to w \to v$ through forward successors w (children of u) and their forward successors v (grandchildren of u, future pole candidates). Then count belly nodes $|\text{children}(u) \cap \text{parents}(v)|$ using the reverse CSR for parents(v). If ≥ 3, the pair forms a Prism candidate with the belly as intermediates. Greedy best-partner selection maximizes belly size per pole.
   Complexity: $O(N_{\text{core}} \cdot D^3)$. With bounded Hasse degree $D \leq 15$: $O(10\text{M} \times 3{,}375) \approx 3.4 \times 10^{10}$ operations at $N = 100\text{M}$ — minutes instead of days.
   Completeness proof: If u and v are poles of a timelike $K_{2,N}$, then every belly node w_i satisfies $u \to w_i \to v$. The forward-forward path $u \to w_i \to v$ guarantees v appears as a candidate of u. Every valid prism partner *must* have at least 3 belly nodes, each providing such a forward-forward path. The 2-hop search is therefore **complete**: no valid partner is missed. The valence guard is safe because any valid pole must have ≥ 3 children (belly nodes).
3. **Threat contraction:** Any external node connected to BOTH poles AND ≥ 2 intermediates is absorbed into the higher-degree pole (vertex contraction).
4. **Generation classification (Vol I phase):** Prisms classified by $\varphi(w) = \text{sign}(\text{bulk\_momentum}[w])$. $g(P) = |\{\varphi(w_i)\}|$ gives the generation number (1, 2, or 3); $\Phi(P) = \sum \varphi(w_i)$ distinguishes matter ($\Phi \geq 0$) from antimatter ($\Phi < 0$).
5. **Transitive merge chain resolution:** Pointer chase to resolve chains ($O(N)$).
6. **Defect CSR construction:** Zero-copy CSR from merge map + pole-completion edges (one edge per Prism: origin $\leftrightarrow$ destination).

4.2 Functions

4.2.1 apply_defect(n, adj_head, adj_data, bulk_momentum) → (DefectResult, TopologySummary, Vec<CausalPrism>)

Main entry point. Finds Causal Prisms ($K_{2,N}$ bipartite structures with belly nodes), classifies them by topological signature into generations, performs threat contraction, and builds the defect CSR by adding pole-completion edges. The committed `Vec<CausalPrism>` is returned alongside the result so that callers (e.g. `--export-slice`) can access the raw prism geometry without re-detection.

4.2.2 process_streaming(n_total, chunk_size, path, seed) → SpectralResult

Out-of-core variant for streaming mode. Single-pass analysis via `scan_edges_with_analysis()`:

1. **Single pass:** Stream edges through `scan_edges_with_analysis()` to simultaneously count degrees, classify core nodes, and collect core-induced edges. Returns `(in_deg, out_deg, is_core, directed_core_edges)`.
2. **Prism detection:** Build CSR for the core-induced subgraph, run $O(N)$ 2-hop Causal Prism detection and generation classification.
3. **Spectral measurement:** Run random walkers on the core CSR for spectral dimension; build directed CSR for causal flux.

4.3 DefectResult Struct

Field	Description
`vac_head`, `vac_data`	Vacuum Hasse CSR (row pointers + column indices).
`def_head`, `def_data`	Defect CSR after threat contraction + Prism pole-completion edges.
`vacuum_core`	Core node indices (top 10% by degree) in the vacuum graph.
`defect_core`	Core node indices surviving threat contraction.
`gen1_nodes`	Generation 1 Prism node indices — most abundant signature (electron-like).
`gen2_nodes`	Generation 2 Prism node indices — second most abundant (muon-like).
`gen3_nodes`	Generation 3 Prism node indices — third most abundant (tau-like).
`anti1_nodes`	Anti-Generation 1 Prism node indices — CPT conjugate (positron-like).
`sterile_nodes`	Sterile prism node indices — prisms with $\Phi = 0$ (fully phase-cancelled, dark matter; Phase-Coherence Theorem).
`merge_into`	Merge map: `merge_into[i]` = canonical node for node i. For threat-absorbed nodes, points to the absorbing Prism pole.

Table 4.1: `DefectResult` fields.

4.4 TopologySummary Struct

Field	Description
total_nodes	Total causet cardinality N.
total_prisms	Number of committed Causal Prisms detected.
max_intermediates	Largest belly size N among all prisms.
count_gen1..3	Number of prisms in each generation.
count_antigen1	Number of antimatter prisms (CPT conjugate of Gen 1).
count_sterile	Number of sterile prisms ($\Phi = 0$, fully phase-cancelled).
avg_mass_gen1..3	Average topological mass $\langle N \rangle$ per generation.
avg_mass_sterile	Average topological mass for sterile prisms.
visible_mass_total	$\sum \lvert\Phi(P)\rvert$ — total visible (EM) mass across all prisms.
dark_mass_total	$\sum(N - \lvert\Phi(P)\rvert)$ — total dark mass across all prisms.
grav_mass_total	$\sum N$ — total gravitational mass (identity: = vis + dark).
omega_ratio	$\Omega_{\text{linear}} = \sum(N - \lvert\Phi\rvert)/\sum\lvert\Phi\rvert$ — linear mass ratio (graph-theoretic identity).
phase_sq_total	$\sum \lvert\Phi(P)\rvert^2$ — total EM self-energy (numerator of $\mathcal{Q}_{\text{topo}}$).
mass_sq_total	$\sum N^2$ — total gravitational self-energy (denominator of $\mathcal{Q}_{\text{topo}}$).
alpha_em	$\alpha = \mathcal{Q}_{\text{topo}}/(8\pi)$ — emergent fine structure constant, where $\mathcal{Q}_{\text{topo}} = \sum\lvert\Phi\rvert^2/\sum N^2$.
omega_energy	$\Omega_{\text{energy}} = 1/\mathcal{Q}_{\text{topo}} - 1$ — self-energy dark matter ratio. Satisfies the exact identity $\alpha(1 + \Omega_{\text{energy}}) = 1/(8\pi)$.
phase_pos_count	Count of intermediate nodes with $\varphi = +1$ (source-like, out-degree > in-degree).
phase_zero_count	Count of intermediate nodes with $\varphi = 0$ (balanced in/out degree).
phase_neg_count	Count of intermediate nodes with $\varphi = -1$ (sink-like, in-degree > out-degree).
prisms_gen1..3	Number of *prisms* (not nodes) per generation.

Table 4.2: TopologySummary fields. The Phase-Coherence Mass Decomposition ($M_{\text{vis}} = |\Phi|$, $M_{\text{dark}} = N - |\Phi|$) replaces the former Conjecture C6 threshold ($N > 5$) with a parameter-free identity. The self-energy ratio Ω_{energy} and the fine structure constant α are algebraically locked by $\alpha(1+\Omega) = 1/(8\pi)$. The phase census fields support the occupancy model analysis (see § 11.4).

4.5 Signature Encoding (Legacy Diagnostic)

Note: The primary generation classification now uses the Vol I phase function ($g(P)$, $\Phi(P)$) described above. The quartile signature encoding below is retained for diagnostic logging only.

Each Causal Prism signature is computed from the bulk momentum of ALL components (origin, destination, and intermediates):

1. Collect bulk momentum $[M_{\text{origin}}, M_{\text{destination}}, M_{w_1}, M_{w_2}, \ldots]$ for all nodes in the Prism.
2. Sort the list and extract quartile values: $[q_0, q_1, q_2, q_3]$ (0th, 25th, 50th, 75th percentiles).
3. Normalise each quartile to $[-10, +10]$ via $q_i \times 10/\max(|q|)$.

4. Offset by $+16$ to make non-negative (range $[6, 26]$, fits in 5 bits).
5. Pack four 5-bit components into a single u32:
$$\mathrm{sig} = c_0 \mid (c_1 \ll 5) \mid (c_2 \ll 10) \mid (c_3 \ll 15)$$

The anti-signature (CPT conjugate) is computed by reversing and negating: $\bar{c}_i = -c_{3-i} + 16$.

CHAPTER 5

Module Reference: `spectral` — Phase 3

Spectral Dimension Measurement

- *Random walk return probability via undirected and directed walkers.*
- *Two tiers: eigendecomposition (exact, $N \leq 3k$ by default; adjustable via `--eigen-cutoff`) and Monte Carlo (above the cutoff).*
- *Per-generation and causal flux measurements.*

5.1 Functions

5.1.1 `spectral_dimension(steps, p_vals)` → `Vec<f64>`

$d_S(t) = -2\,d(\ln P)/d(\ln t)$ via centred finite differences. Forward difference at $t = 1$, backward at $t = t_{\max}$, centred elsewhere.

5.1.2 `run_walkers(adj_head, adj_data, starts, steps, seed)` → `Vec<f64>`

Run W independent lazy-walk walkers in parallel. At each step: 50% stay, 50% move to a uniformly random neighbour. Returns $P(t)$ = fraction of walkers that returned to their starting node at each step. Parallel via Rayon.

Strict Finitism: all internal accumulation uses u64 integer arithmetic. Each walker contributes exactly 0 or 1 (Kronecker delta) per measurement step. The reduction sums u64 counts across all walkers. The single `f64` division `count / W` occurs only at the final return, preserving exact integer semantics throughout the Monte Carlo core.

5.1.3 `run_transmission_walkers(…)` → `(Vec<f64>, Vec<f64>)`

Directed walkers (mandatory move along causal arrow). Returns two vectors: attraction flux (Gen1 → AntiGen1) and repulsion flux (Gen1 → Gen1). Walkers that reach a dead end (edge of causal set) escape and stop. Same u64 integer accumulation as `run_walkers`.

5.1.4 `compute_eigen(n, rows, cols, steps, core)` → `SpectralResult`

Exact $P(t) = N^{-1} \sum_k \lambda_k^t$ from eigendecomposition of the symmetric normalised adjacency $S = D^{-1/2} A D^{-1/2}$. Complexity: $O(N^3)$. Used only for small graphs ($N \leq 3$k). Local $P(t)$ uses eigenvector-weighted sums over core indices.

5.1.5 `compute_monte_carlo(n, rows, cols, steps, core, n_w, rng)` → `SpectralResult`

Monte Carlo $P(t)$ from random walkers. Builds CSR adjacency internally, then launches lazy random walkers. Complexity: $O(W \cdot t)$, independent of N.

5.1.6 `compute_monte_carlo_csr(n, head, data, steps, core, n_w, rng)` → `SpectralResult`

Zero-copy variant — avoids rebuilding adjacency when CSR is already available from Phase 2. This is the hot path for large N in-memory runs.

5.1.7 `build_adj_list(n, rows, cols)` → `(Vec<u32>, Vec<u32>)`

Build sorted undirected adjacency list in CSR format from edge lists.

5.1.8 `make_symmetric(n, head, data)` → `(Vec<u32>, Vec<u32>)`

Convert a directed CSR (forward-only Hasse DAG) into an undirected adjacency list. Used to symmetrize the vacuum CSR before running spectral walkers, since the Hasse diagram stores only forward (past→future) edges but random walkers need to step in both directions.

5.2 SpectralResult Struct

Field	**Description**
`p_global`	$P(t)$ from uniformly random starts. Probes global geometry.
`ds_global`	$d_S(t)$ of the bulk (derived from `p_global`).
`p_local`	$P(t)$ from walkers starting on core nodes.
`ds_local`	$d_S(t)$ from core walkers. Sensitive to topological defects.
`p_gen1`, `ds_gen1`	$P(t)$ and d_S for Generation 1 Causal Prism nodes (electron-like).
`p_gen2`, `ds_gen2`	$P(t)$ and d_S for Generation 2 Prism nodes (muon-like).
`p_gen3`, `ds_gen3`	$P(t)$ and d_S for Generation 3 Prism nodes (tau-like).
`p_anti1`, `ds_anti1`	$P(t)$ and d_S for Anti-Generation 1 (positron-like, CPT conjugate).
`flux_attraction`	Directed transmission: Gen1 → AntiGen1 (opposite charge).
`flux_repulsion`	Directed transmission: Gen1 → Gen1 (same charge).
`flux_attr_norm`	Normalized attraction flux: `flux_attraction` / |targets|. Isolates intrinsic per-node coupling strength (per-charge definition, Vol II).
`flux_repu_norm`	Normalized repulsion flux: `flux_repulsion` / |targets|.
`p_sterile`, `ds_sterile`	$P(t)$ and d_S for walkers on sterile prism nodes ($\Phi = 0$). Phase-Coherence Theorem: gravitationally active, electromagnetically silent (dark matter).
`ds_*_std`	Population standard deviation of each $d_S(t)$ field across M realisations. Empty when $M = 1$. Provides error bars for publication figures.
`flux_*_std`	Population standard deviation of each flux field across M realisations.

Table 5.1: `SpectralResult` fields (28 total).

CHAPTER 6

Module Reference: `anim_export` — Topology Slice Export

Binary Export for External Renderers

- *Pre-computed binary topology data for keyframe playback (e.g. CausalAnim).*
- *Invoked via `--export-slice <PATH>`: runs Phases 1–2 only, no spectral walkers.*
- *Output format: bincode-serialized `TopologySlice` containing coordinates, directed Hasse edges, and Causal Prisms.*

6.1 Structs

6.1.1 `TopologySlice`

The top-level serializable structure written to the `.slice` file:

Field	Description
`n_total`	Number of spacetime events.
`coordinates`	`Vec<[f64; 4]>` — 4D coordinates (time-sorted) for all events.
`hasse_edges`	`Vec<(u32, u32)>` — directed Hasse links (past→future), filtered by time ordering $t_u < t_v$.
`prisms`	`Vec<PrismDef>` — all committed Causal Prisms from Phase 2.

6.1.2 `PrismDef`

Lightweight `u32`-indexed Causal Prism definition (converted from internal `CausalPrism` with `usize` indices):

Field	Description
`origin`	Past pole index (u32).
`destination`	Future pole index (u32).
`intermediates`	`Vec<u32>` — belly node indices.

6.2 Function

6.2.1 `write_slice(path, slice)` → `Result<(), String>`

Serializes a `TopologySlice` to bincode format and writes the bytes to the given file path. At $N = 50{,}000$ the output file is typically ~5 MB; at $N = 5{,}000$ it is ~800 KB.

6.3 Usage

The `--export-slice` flag triggers an early-exit branch in `main.rs`:

1. Run Phase 1: `sprinkle` + `build_hasse_sparse` or `build_hasse_direct`.
2. Run Phase 2: `apply_defect` → `DefectResult` + `Vec<CausalPrism>`.
3. Extract directed Hasse edges from the vacuum CSR (filter by $t_u < t_v$).

4. Convert `CausalPrism` → `PrismDef` (`usize` → `u32`).
5. Serialize `TopologySlice` via `write_slice` and exit.

Phases 3 and 4 (spectral walkers, ensemble averaging, CSV output) are skipped entirely, making the export fast even for large N.

CHAPTER 7

Module Reference: `output` — Phase 4

CSV Serialisation

- *Ensemble-averaged observables written to CSV (`results.csv`).*
- *32-column format: return probabilities, spectral dimensions, raw and normalized causal flux, sterile prism observables ($\Phi = 0$), mass spectrum, and ensemble standard deviations (error bars for $M > 1$ runs).*
- *Topology summary (`topology_summary.csv`): prism census, generation abundances, Phase-Coherence Mass Decomposition (Ω_d/Ω_v, α).*
- *Mass spectrum histogram (`mass_spectrum.csv`): belly size N vs frequency.*

7.1 Functions

7.1.1 `write_csv(path, steps, vacuum, defect, metadata)`

Serialise vacuum and defect spectral results to a CSV file. Each row corresponds to one diffusion step t. The vacuum result provides control measurements (unmodified Hasse graph); the defect result provides measurements after Kuratowski contraction (Causal Prism + K_5 threat absorption).

7.1.2 `export_topology_summary(path, topo, metadata)`

Export a single-record key-value CSV containing the structural fingerprint of the generated universe: prism counts, generation abundances, average topological masses, and the Phase-Coherence Mass Decomposition fields (`visible_mass_total`, `dark_mass_total`, `grav_mass_total`, `omega_ratio`, `phase_sq_total`, `mass_sq_total`, `q_topo`, `alpha_em`, `omega_energy`).

7.1.3 `export_mass_spectrum(path, histogram, metadata)`

Export the prism mass spectrum histogram as a two-column CSV (`intermediates_N`, `frequency`). Each row maps a belly size N to the number of committed prisms with that exact belly size.

7.2 CSV Column Reference

Column	Physics	Units
step	Diffusion time t	integer steps
P_vac	Return probability on vacuum graph	probability
dS_vac	Spectral dimension of vacuum	dimensionless
P_def	Return probability on defect graph	probability
dS_def	Spectral dimension with matter	dimensionless
P_Gen1	Return probability for Generation 1	probability
dS_Gen1	Spectral dimension, Generation 1	dimensionless
P_Gen2	Return probability for Generation 2	probability
dS_Gen2	Spectral dimension, Generation 2	dimensionless
P_Gen3	Return probability for Generation 3	probability
dS_Gen3	Spectral dimension, Generation 3	dimensionless
P_Anti1	Return probability for antimatter	probability
dS_Anti1	Spectral dimension of antimatter	dimensionless
Flux_Attr	Causal flux: Gen1→AntiGen1 (attraction)	probability
Flux_Repu	Causal flux: Gen1→Gen1 (repulsion)	probability
Flux_Attr_Norm	Normalized flux: Attr / |targets| (per-charge def.)	per-node prob
Flux_Repu_Norm	Normalized flux: Repu / |targets| (per-charge def.)	per-node prob
P_Sterile	Return prob. for sterile prisms ($\Phi = 0$)	probability
dS_Sterile	Spectral dimension of sterile prisms (Phase-Coherence Thm)	dimensionless
Mass_Gen1..3	Avg topological mass per generation	integer (avg)
Mass_Anti1	Avg topological mass for antimatter	integer (avg)
dS_vac_std	Std dev of d_S vacuum across realisations	dimensionless
dS_def_std	Std dev of d_S defect across realisations	dimensionless
dS_Gen1..3_std	Std dev of d_S per generation	dimensionless
dS_Anti1_std	Std dev of d_S antimatter	dimensionless
dS_Sterile_std	Std dev of d_S sterile	dimensionless
Flux_Attr_std	Std dev of flux attraction	probability
Flux_Repu_std	Std dev of flux repulsion	probability

Table 7.1: CSV output columns (32 total).

CHAPTER 8

Module Reference: `measurement` — Observer Modules

Phase 3.5: Four Observer Measurements

- *Read-only algorithms that extract physics from existing simulation data without modifying the underlying engine.*
- *Enabled via `--measure-all` or per-module flags (`--measure-mass`, `--measure-halflife`, `--measure-modulo`, `--measure-vacuum`).*
- *Output CSVs: `traversal_mass.csv`, `half_life.csv`, `modulo_interference.csv`, `vacuum_polarization.csv`.*

8.1 M1 — Cover-Time Mass (Topological Inertia)

The dynamical mass of a Causal Prism is measured via the **coupon-collector cover time**: a confined random walker on the defect CSR must visit *every* belly node before exiting at the destination pole.

8.1.1 The $K_{2,N}$ Hitting-Time Theorem

On a bipartite complete graph $K_{2,N}$, all N intermediates are structurally equivalent. The hitting time from origin to destination satisfies:

$$E[T \mid \text{origin}] = 1 + E[T \mid \text{int}] \tag{8.1}$$
$$E[T \mid \text{int}] = 1 + \tfrac{1}{2} \cdot 0 + \tfrac{1}{2} \cdot E[T \mid \text{origin}] \tag{8.2}$$

Solving: $E[T] = 4$ (or ~ 8 with lazy walk), *independent of* N. Adding more intermediates gives more choices at the origin, but each intermediate still offers the same $\frac{1}{2}$ chance of reaching the destination.

Consequence: simple origin→destination traversal cannot measure mass.

8.1.2 Cover Time as Dynamical Mass

For a particle to fully propagate, the causal flux must update the *entire* internal structure. The cover time—the number of steps until the walker has visited every belly node—scales as $O(N \log N)$ via the coupon-collector problem:

$$E[T_{\text{cover}}] = N \cdot H_N \approx N \ln N + \gamma N$$

where H_N is the N-th harmonic number and $\gamma \approx 0.577$ is the Euler–Mascheroni constant. If the walker reaches the destination before full coverage, it reflects back to the origin (incomplete phase update, physically analogous to causal flux bouncing inside the particle).

8.1.3 Results (N=5000, M=8)

Generation	Mean Cover Time	Traversals	Ratio
Gen1	1197	16,856	1.000
Gen2	1145	371,123	0.957
Gen3	1417	71,001	**1.184**

Table 8.1: Cover-time mass ratios. Gen3/Gen1 = 1.18 reflects the static mass spectrum (Gen3 belly = 8.31 vs Gen1 belly = 6.46). The Gen2/Gen1 inversion is a finite-size artifact: Gen1 prisms are rare (22 per realization), and the belly difference (6.81 vs 6.46) is within coupon-collector variance.

8.2 M2 — Half-Life Census

Cross-ensemble stability statistics. For each prism, records generation, belly size N, and net phase Φ. Computes:

- Occupancy fractions $p(g_k \mid N)$ per belly size.
- Stability ratios $\tau(\text{gen2})/\tau(\text{gen1})$ and $\tau(\text{gen3})/\tau(\text{gen1})$.

The occupancy fractions at $N = 5000$, $M = 8$ are: Gen1 = 4%, Gen2 = 79%, Gen3 = 17%, matching the zero-parameter occupancy model prediction (2% / 85% / 13%) from the coupon-collector selection effect on the phase trichotomy $(p_+, p_0, p_-) = (0.318, 0.018, 0.664)$.

8.3 M3 — Modulo Path Integral

NTT-based interference fringes from walker phases. Each walker carries a modular phase $\phi = g^S \bmod p$ where $g = 3$ is a primitive root and $p = 65537$ is the Fermat prime F_4. The exponent S counts cumulative *actual* moves (not stays).

Per-node phase accumulation uses `AtomicU64` (Relaxed ordering) for lock-free parallel writes. Post-processing: symmetric representation $c = \min(\phi, p - \phi)$, intensity $I = c^2/(p/2)^2$, classification into constructive $(I > \mu + 2\sigma)$ and destructive $(I < \mu - 2\sigma)$ nodes.

Requires $N \geq 50{,}000$ for meaningful spatial resolution.

8.4 M4 — Vacuum Polarization

$K_{3,3}$ screening at Gen1 prism free ports. For each Gen1 prism P (a $K_{2,3}$):

1. Gather candidate nodes: all neighbours of P's 5 nodes (2 poles + 3 intermediates) in both forward and reverse CSR, excluding existing prism members.
2. For each candidate C: check whether adding edges $C \leftrightarrow$ poles would create a $K_{3,3}$ subdivision. The $K_{3,3}$ threat fires when C connects to both poles AND ≥ 3 intermediates.
3. Count `n_attempted`, `n_rejected_k33`, `n_accepted`.
4. Local screening $= n_{\text{accepted}}/n_{\text{attempted}}$.

At $N = 5000$: zero screening (vacuum too sparse for $K_{3,3}$ formation). As $N \to \infty$, screening increases and α^{-1} flows from the bare value (~ 165) toward the physical 137.

8.5 CLI Flags

Flag	Default	Effect
`--measure-mass`	off	Enable cover-time mass (M1)
`--measure-halflife`	off	Enable half-life census (M2)
`--measure-modulo`	off	Enable modulo path integral (M3)
`--measure-vacuum`	off	Enable vacuum polarization (M4)
`--measure-all`	off	Enable all four measurements
`--modulo-prime`	65537	Prime modulus for M3
`--modulo-root`	3	Primitive root for M3

Table 8.2: Measurement module CLI flags.

CHAPTER 9

Module Reference: `memory` — RAM-Aware Mode Selection

Chapter Overview

- *Automatic detection of available system RAM.*
- *Memory estimation heuristics for each simulation tier.*
- *Interactive mode selection with config file override.*

9.1 Functions

9.1.1 `load_config(dir)` → `Config`

Load configuration from a TOML file, or return defaults. Searches for `causal_set.toml` in the given directory first, then falls back to the current working directory.

9.1.2 `ensure_config(dir)`

Write the default config file if it doesn't already exist.

9.1.3 `recommend_mode(n, cfg)` → `(ExecMode, u64, u64)`

Recommend an execution mode based on config + available RAM vs estimated usage. The estimate accounts for concurrent realisations (default: 2 for $N > 10$M, 4 otherwise; overridable via `--threads`).

9.1.4 `prompt_mode(recommended, available, estimated, cfg)` → `ExecMode`

Prompt the user to confirm or override the recommended execution mode.

9.1.5 `estimate_memory_bytes(n)` → `u64`

Heuristic peak memory estimate for in-memory mode:

- $N \leq 3{,}000$: $\sim N^2 \times 24$ bytes (dense eigendecomp matrices).
- $3{,}000 < N \leq 50{,}000$: $\sim N \times 2{,}000$ bytes (sparse structures).
- $N > 50{,}000$: $\sim N \times 500$ bytes (OCI + CSR + defect).

9.1.6 `estimate_streaming_memory_bytes(n)` → `u64`

Peak memory estimate for streaming mode: $13N + 100\,\text{MB}$. Components: $8N$ (degree arrays) $+50\cdot(N/10)$ (core CSR) $+100\,\text{MB}$ (buffer). At $N = 100$M: $\sim$1.3 GB per realisation.

9.1.7 `max_concurrent_runs(n)` → `usize`

Default concurrency heuristic: 2 for $N > 10$M, 4 otherwise. Used as the fallback when `--threads` is not provided.

9.1.8 `available_ram_bytes()` → `u64`

Detect available system RAM in bytes via the `sysinfo` crate.

9.2 Types

Definition 1: ExecMode

An enum: `InMemory` (keep all data in RAM) or `Streaming` (single-pass edge analysis via `scan_edges_with_analysis()`, ~1.3 GB per realisation at $N = 100\text{M}$).

Definition 2: Config

User-configurable settings: `max_ram_bytes` (hard ceiling, $0 = $ auto), `safety_fraction` (0.0–1.0), `output_dir` (default output directory).

CHAPTER 10

Usage

10.1 Options

Argument	Description	Default
`N`	Number of spacetime events	5000
`M`	Ensemble realisations	10
`output_dir`	Directory for results	current dir
`--stream`	Force streaming mode	—
`--inmemory`	Force in-memory mode	—
`--seed` *u64*	Base seed (each realisation uses seed+i)	system clock
`--threads` *usize*	Max parallel realisations	auto (2–8)
`--epsilon` *f64*	Topological error tolerance ε	0.01
`--tmax` *usize*	Maximum diffusion time	15
`--eigen-cutoff` N	Eigendecomp threshold	3000
`--target-error` *f64*	Relative SE threshold for adaptive convergence	0.05
`--max-ensemble` *usize*	Maximum realisations (hard cap)	22
`--min-ensemble` *usize*	Minimum realisations before convergence	3
`--batch-size` *usize*	Realisations per parallel batch	auto
`--resume`	Resume from checkpoint	—
`--export-slice` *PATH*	Export topology slice (single realization, no ensemble)	—
`--help`	Show help message	—

10.2 Configuration

Place a `causal_set.toml` in the working directory (auto-generated on first run if absent):

```
# Hard limit on RAM (GB). 0 = auto-detect from system.
max_ram_gb = 0

# Fraction of available RAM considered safe (when auto-detecting).
safety_fraction = 0.70

# Default output directory for results and streaming edge files.
output_dir = "."
```

10.3 Examples

```
# Standard in-memory run
cargo run --release -- 50000 10

# Streaming run (large N, single-pass O(N))
cargo run --release -- 1000000 5 --stream

# Reproducible run with explicit seed
cargo run --release -- 50000 10 --inmemory --seed 12345

# 16 parallel realisations on a high-RAM workstation
```

```
cargo run --release -- 10000000 20 --inmemory --threads 16

# Force eigendecomp up to N=5000 for exact comparison
cargo run --release -- 5000 10 --inmemory --eigen-cutoff 5000

# Custom walker precision (fewer walkers for exploratory runs)
cargo run --release -- 50000 10 --inmemory --epsilon 0.05 --tmax 10

# Custom output directory
cargo run --release -- 50000 10 /path/to/output

# Export topology slice for external renderers (Phases 1-2 only)
cargo run --release -- 50000 --inmemory --seed 42 \
  --export-slice /tmp/universe.slice
```

CHAPTER 11

Expected Results

11.1 Spectral Dimension

Production ensemble: $N = 10^7$, $M = 20$ realisations, seed 42, runtime ∼4.5 hours on a ThinkPad T480 (Intel i5-8250U, 16 GB RAM).

- **Vacuum (UV, σ=1):** $d_S = 1.949 \pm 0.003$ (UV dimensional reduction $d_S \to 2$).
- **Vacuum (IR, σ=4):** $d_S = 4.956 \pm 0.001$ (4D Minkowski continuum emerges).
- **Defect (core, peak):** $d_S \approx 9.06$ (topological retention — the Prism bipartite structure traps probability flow and elevates the effective dimension).

Key Result

Mass is not a continuous parameter assigned to a particle, but a quantized topological property: the integer count N of intermediate nodes in the Causal Prism. This topological inertia stabilises the spectral dimension at a finite value, while the vacuum diverges.

11.2 Generation Classification

Causal Prisms are classified by the Vol I phase function $\varphi(w) = \text{sign}(\text{bulk\_momentum}[w])$:

Generation	Physics Analogue	Classification	Mass (N=10^7, M=20)
Gen1	Electron-like	g=1 ∧ Φ≥0	4.555 (lightest)
Gen2	Muon-like	g=2	6.531 (medium)
Gen3	Tau-like	g=3	7.732 (heaviest)
AntiGen1	Positron-like	g=1 ∧ Φ<0	4.45 (CPT, $\Delta m/m = 2.3\%$)

At $N = 10^7$ (M=20): Total prisms: 7,723,943. g=1: 63,278 (2.1%), g=2: 2,552,871 (84.9%), g=3: 391,039 (13.0%). Sterile (Φ=0): 400,986. Mass ratios 1 : 1.43 : 1.70 are N-independent topological invariants.

11.3 Coupling Constants

Observable	N=10^7	$N\to\infty$	Note
$\mathcal{Q}_{\text{topo}}$	0.191	0.152 ± 0.001 (R^2=0.9996)	Topological charge ratio
α^{-1}	131.8	165.1 ± 1.0	Bare coupling at Planck scale
Ω_{energy}	4.24	5.57 (Planck 2018: 5.36)	Self-energy dark matter ratio
$\alpha(1+\Omega)$	$1/(8\pi)$	$1/(8\pi)$	Exact at every N

11.4 Occupancy Model (Analytical Mass Hierarchy)

The mass hierarchy $m_1 < m_2 < m_3$ is explained analytically as a **coupon-collector selection effect** on the belly size distribution $f(N)$. Each intermediate node w carries a causal phase $\varphi(w) = \text{sign}(\text{out\_degree} - \text{in\_degree}) \in \{-1, 0, +1\}$. The generation $g(P)$ counts how many distinct phase signs appear among the N intermediates of prism P.

Since $P(g{=}1 \mid N) \sim p_{\max}^N$ decays exponentially, Gen 1 prisms are biased toward small bellies (low mass), while Gen 3 prisms require all three signs (large bellies, high mass). The expected mass per generation is

$$E[N \mid g{=}k] = \frac{\sum_N N \cdot P(g{=}k \mid N) \cdot f(N)}{\sum_N P(g{=}k \mid N) \cdot f(N)},$$

where the generation probabilities follow from inclusion-exclusion:

$$P(g{=}1 \mid N) = p_+^N + p_0^N + p_-^N,$$
$$P(g{=}3 \mid N) = 1 - (1{-}p_+)^N - (1{-}p_0)^N - (1{-}p_-)^N + p_+^N + p_0^N + p_-^N.$$

The simulation reports the **intermediate phase census**: $(p_+, p_0, p_-) = (0.318,\ 0.018,\ 0.664)$. Plugging these measured fractions into the occupancy formula reproduces the mass ratios with **zero free parameters**:

Ratio	Predicted	Observed	Error
m_2/m_1	1.409	1.434	1.8%
m_3/m_1	1.674	1.698	1.4%

Table 11.1: Occupancy model predictions versus simulation. Both inputs—the belly distribution $f(N)$ and the phase fractions—are determined entirely by the Poisson sprinkling geometry. Analysis script: `data/scripts/occupancy_model.py`.

11.5 Causal Flux

Directed walker transmission probabilities measure electromagnetic-like interactions:

- **Attraction:** Gen1 → AntiGen1 (opposite charge, positive flux).
- **Repulsion:** Gen1 → Gen1 (same charge, lower transmission).

CHAPTER 12

Dependencies

Crate	Purpose
`nalgebra`	Dense eigendecomposition (Tier 1, $N \leq$ cutoff; default 3k)
`sprs`	Sparse matrix algebra for A^2 Hasse reduction
`rand`	Deterministic PRNG (seeded `StdRng`)
`rayon`	Data-parallel iteration across walkers and nodes
`sysinfo`	System RAM detection for automatic mode selection
`smallvec`	Stack-allocated small vectors for inner-loop optimisation
`serde` + `bincode`	Serialization for checkpoint files and `.slice` topology export

Table 12.1: Rust crate dependencies.

See Also

- *Modulo Synthesis Vol I: The Geometry of Time* — Axiom 1, causal sets, spectral dimension flow, the Geometry of Logic ($O(N)$ Locality Theorem), Law of Large Numbers.
- *Modulo Synthesis Vol II: The Geometry of Matter* — Axiom 2, Kuratowski's Theorem, generation theorem, colour confinement, gauge forces, the Three Geometries.
- *Emma's Thought Experiments* — Pedagogic companion with intuitive explanations, dimensional adjustment, and locality.

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"The universe is strictly computable, finite, and structurally homeostatic.
The rest is just counting."